

RMO AI Visual Impairment Indicators: Test Report

Test date: 28 Sep 2026 · System version: visual-indicators-rules-v1 (deployed AI service container, rebuilt 28 Sep 2026 16:36 IST) · Prepared by: Engineering

10

video clips tested

11.5 min

footage analysed (3,453 frames)

1 / 5

impairment clips raised an alert

0 / 5

false alerts on normal clips

Executive summary

- **The pipeline works end to end.** The deployed AI service finds people in video, tracks them, reads body posture and facial signals, and produces a live status (Normal / Monitoring / Elevated / High / Insufficient evidence) with written evidence for each person.
- **It is cautious by design.** Across 5 normal-behaviour clips it raised **no false alerts**. It raised an alert on **1 of 5** impairment clips: a staged 1916 film showing exaggerated drunk behaviour, where it reached **High**, with the first alert 50 seconds into the clip.
- **It did not alert on any of the 3 real-world impairment clips** (police dash-cam and body-cam). On 2 of them it showed the early-warning *Monitoring* status. The main reasons were camera conditions: face or lower body out of view, night footage, and subjects mostly standing still or seated.
- **Results are reproducible.** A second, independent run on a different machine gave the same highest status on 9 of 10 clips. The tenth differed only between *Insufficient evidence* and *Normal*.
- **Software checks pass:** 74 of 74 AI-service unit tests, 41 of 41 platform API tests (safety events, review, audit, access control, live alerts, licensing) and a clean type-check.
- **Recommendation:** the system is ready for a **supervised pilot with data collection** in 1–2 lobbies. It is **not** ready to drive any automatic action. Alerts must stay a prompt for a human check (e.g. breath test).

Important: the system reports *visual indicators* such as unsteady posture, sway or eye closure. It does not diagnose intoxication. Illness, fatigue, injury or disability can produce the same indicators. Every alert goes to a trained reviewer.

What was tested and how

- **Footage:** 10 clips (11.5 minutes, about 3,450 frames) from Wikimedia Commons, all public domain or openly licensed. There are 5 impairment clips (2 staged films, 3 real police recordings) and 5 normal clips (people walking or standing).
- **Method:** each clip was replayed frame by frame at 5 frames per second through the **same AI service container that runs in the application**, using the video's own timing, the same way a live lobby call is processed.
- **Measured:** the highest status reached, how long each status was shown, when the first alert appeared, image quality, how often the face and full body were visible, and which evidence the system reported.
- **Pass criteria:** an impairment clip passes if it reaches *Elevated* or *High*. A normal clip passes if it never reaches *Elevated* or *High*. *Monitoring* is an early-warning state and does not raise an alert.

How the system decides

Six signal groups are scored per person over a rolling 10-second window: gait, sway, posture, head movement, eyes and arm coordination. An **Elevated** alert needs a combined score of at least 0.5, with at least two groups independently strong, sustained for 8 seconds. **High** needs at least 0.7 sustained for 10 seconds. The system abstains (*Insufficient evidence*) when the view is too poor or too short. These rules deliberately trade sensitivity for fewer false accusations.

Results per clip

Clip	Type	Length	Expected	Highest status reached	First alert	Result
Dream of a Rarebit Fiend (Edwin Stanton Porter)	Staged drunk (actor, 1906 film)	75s	Alert	Monitoring	—	Missed
One A.M. (1916) by Charlie Chaplin	Staged drunk (Chaplin, 1916 film)	270s	Alert	High	50s	Detected
Suspected Drunken Driver Confused About Being Arrested Before Proceeding to Mess Up The Alphabet	Real DUI suspect — roadside test, night dashcam	46s	Alert	Normal	—	Missed
Suspected Drunken Driver Confused About Being Arrested Before Proceeding to Mess Up The Alphabet	Real DUI suspect — standing in booking room	47s	Alert	Monitoring	—	Missed
Fake Cop Handcuffs Drunk Man, Gets Arrested Himself by Atlanta Officers	Real intoxicated man — shaky police bodycam	160s	Alert	Monitoring	—	Missed
Inmate walking in shackles	Sober walking, ankle shackles (look-alike)	25s	No alert	Insufficient evidence	—	Correct (no alert)
Walking in the sands	Sober walking on sand (look-alike)	18s	No alert	Monitoring	—	Correct (no alert)
Kobuleti2025ispani-walking	Sober walking, street (vertical video)	38s	No alert	Normal	—	Correct (no alert)
Deb Haaland and Isleta Pueblo Chief Judge Verna Teller walking at the Capitol	Sober walking, 6 s clip	6s	No alert	Normal	—	Correct (no alert)

Kamala Harris and Joe Biden walking at the White House in slow motion	Sober walking, slow motion, 5 s	5s	No alert	Normal	—	Correct (no alert)
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Clip details

Each timeline shows one row per second: the coloured strip is the status the system displayed, the blue line is the combined risk score (0–1). Dashed lines mark the Elevated (0.5) and High (0.7) score thresholds.

■ No person ■ Insufficient evidence ■ Normal ■ Monitoring ■ Elevated ■ High

Dream of a Rarebit Fiend (Edwin Stanton Porter) — Missed

Staged drunk (actor, 1906 film) · Public domain · [source](#)



Highest status: Monitoring

Time in each status: Insufficient evidence 33s, No person 31s, Normal 6s, Monitoring 5s

Image quality (0–1): 0.39 · **Face visible:** 1% · **Full body visible:** 87%

Strongest signals: sway 1.00, posture 0.96, gait 0.91, coordination 0.69

Main evidence shown: Posture instability, Irregular gait, Lateral sway while standing

Main limitations shown: Face not visible, Too few visible signals, Low video quality

Posture, sway and gait signals were strong, but image quality was low (0.39): old, grainy film, the actor's face visible in only 1% of frames, and many camera cuts. The combined evidence never stayed high for the full 8 seconds, so it stopped at Monitoring.



One A.M. (1916) by Charlie Chaplin — Detected

Staged drunk (Chaplin, 1916 film) · Public domain · [source](#)



Highest status: High

Time in each status: Monitoring 132s, Elevated 47s, No person 43s, Insufficient evidence 25s, Normal 22s, High 1s

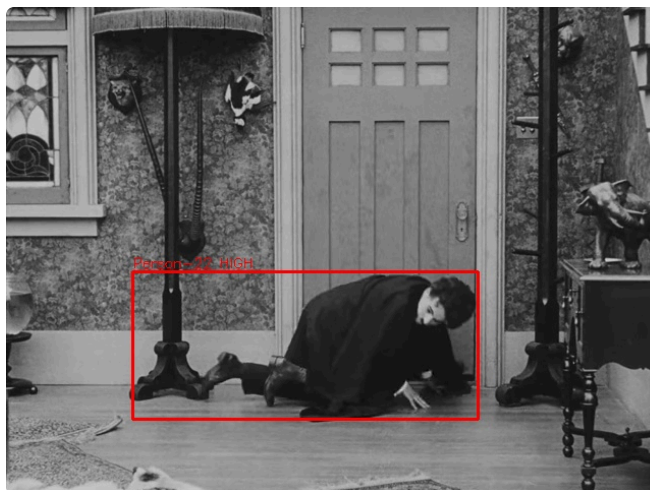
Image quality (0–1): 0.66 · **Face visible:** 48% · **Full body visible:** 60%

Strongest signals: head 1.00, sway 1.00, posture 1.00, gait 0.91, coordination 0.84, eyes 0.71

Main evidence shown: Head instability, Posture instability, Lateral sway while standing

Main limitations shown: Lower body not visible — gait unavailable, Face not visible, Too few visible signals

The actor's exaggerated staggering, swaying and falling produced strong posture, sway and head signals. The system raised 7 alert episodes, 6 Elevated and 1 High (46.6 s Elevated, 1.2 s High in total). Scene cuts in the film split the one actor into several tracked persons, so one person can produce several episodes. Live lobby cameras have no scene cuts.



Suspected Drunken Driver Confused About Being Arrested Before Proceeding to Mess Up The Alphabet — Missed

Real DUI suspect — roadside test, night dashcam · Public domain · [source](#)

0.7
0.5

46s

Highest status: Normal

Time in each status: No person 42s, Insufficient evidence 3s, Normal 2s

Snapshot omitted: real, identifiable person.

Image quality (0–1): 0.39 · **Face visible:** 0% · **Full body visible:** 42%

Strongest signals: posture 0.36, sway 0.35, coordination 0.21

Main evidence shown: —

Main limitations shown: Face not visible, Too few visible signals, Lower body not visible — gait unavailable

Night dash-cam with the subject small and far away. There was no person in view for most of the segment (42 of 46 s). While the subject was visible, the system mostly abstained (Insufficient evidence). In the last 2 seconds it showed Normal, based on weak signals.

Suspected Drunken Driver Confused About Being Arrested Before Proceeding to Mess Up The Alphabet — Missed

Real DUI suspect — standing in booking room · Public domain · [source](#)



Highest status: Monitoring

Time in each status: Monitoring 24s, Normal 17s, Insufficient evidence 4s, No person 2s

Snapshot omitted: real, identifiable person.

Image quality (0–1): 0.66 · **Face visible:** 100% ·

Full body visible: 0%

Strongest signals: eyes 1.00, sway 1.00, head 0.71, coordination 0.29, posture 0.00

Main evidence shown: Prolonged eye closure, Head instability, Lateral sway while standing

Main limitations shown: Lower body not visible — gait unavailable, Too few visible signals

This is the closest real case. The suspect is standing in a booking room with eyes closed and head rolling: the eye signal reached 1.0, head 0.71 and sway 1.0. The combined score peaked at 0.495, just under the 0.5 Elevated threshold, so it stayed at Monitoring. His legs are out of frame, so gait could not be measured.

Fake Cop Handcuffs Drunk Man, Gets Arrested Himself by Atlanta Officers — Missed

Real intoxicated man — shaky police bodycam · Public domain · [source](#)

160s



Highest status: Monitoring

Snapshot omitted: real, identifiable person.

Time in each status: No person 74s, Insufficient evidence 57s, Monitoring 19s, Normal 10s

Image quality (0–1): 0.56 · **Face visible:** 31% · **Full body visible:** 10%

Strongest signals: head 1.00, coordination 0.62, sway 0.35, posture 0.10, eyes 0.00

Main evidence shown: Head instability, Jerky arm movements / repeated corrections

Main limitations shown: Lower body not visible — gait unavailable, Too few visible signals, Face not visible

Shaky body-cam footage with many people (32 tracks). The subject's lower body was mostly out of view. Head and arm signals triggered Monitoring only.

Inmate walking in shackles — Correct (no alert)

Sober walking, ankle shackles (look-alike) · CC0 · [source](#)

26s

0.7
0.5

Highest status: Insufficient evidence

Time in each status: Insufficient evidence 25s, No person 1s

Image quality (0–1): 0.56 · **Face visible:** 0% · **Full body visible:** 0%

Strongest signals: —

Main evidence shown: —

Main limitations shown: Face not visible, Lower body not visible — gait unavailable, Too few visible signals

Face and lower body were mostly not visible, so the system correctly abstained (Insufficient evidence) rather than guessing.

Snapshot omitted: real, identifiable person.

Walking in the sands — Correct (no alert)

Sober walking on sand (look-alike) · CC BY 3.0 · [source](#)

18s

0.7
0.5

Highest status: Monitoring

Time in each status: Monitoring 8s, Normal 6s, Insufficient evidence 4s

Image quality (0–1): 0.49 · **Face visible:** 1% · **Full body visible:** 100%

Strongest signals: sway 1.00, gait 0.50, coordination 0.22, posture 0.00

Main evidence shown: Lateral sway while standing, Irregular gait

Main limitations shown: Face not visible, Too few visible signals

The one normal clip that reached Monitoring (7.8 s): a person walking slowly away from the camera was partly read as swaying on the spot. It never reached an alert. This is a known limitation, listed below.

Snapshot omitted: real, identifiable person.

Kobuleti2025ispani-walking — Correct (no alert)

Sober walking, street (vertical video) · CC BY 4.0 · [source](#)

38s

0.7
0.5

Highest status: **Normal**

Time in each status: Normal 16s, No person 15s,
Insufficient evidence 6s

Snapshot omitted: real, identifiable person.

Image quality (0–1): 0.54 · **Face visible:** 0% · **Full
body visible:** 100%

Strongest signals: sway 0.36, coordination 0.05,
posture 0.00

Main evidence shown: —

Main limitations shown: Face not visible, Too few
visible signals

Before the fixes described below, this normal walk
was shown as Monitoring for 13 s. It now stays
Normal.

Deb Haaland and Isleta Pueblo Chief Judge Verna Teller walking at the Capitol — **Correct (no alert)**

Sober walking, 6 s clip · Public domain · [source](#)



Highest status: **Normal**

Time in each status: Insufficient evidence 4s,
Normal 2s

Snapshot omitted: real, identifiable person.

Image quality (0–1): 0.63 · **Face visible:** 51% · **Full body visible:** 96%

Strongest signals: sway 1.00, head 0.70,
coordination 0.24, posture 0.00, eyes 0.00

Main evidence shown: Head instability, Lateral
sway while standing

Main limitations shown: Face not visible, Too few
visible signals, Lower body not visible — gait
unavailable

Short clip, with only about 6 s of usable footage. Its
result carries little weight.

Kamala Harris and Joe Biden walking at the White House in slow motion — Correct (no alert)

Sober walking, slow motion, 5 s · Public domain · [source](#)

5s

0.7
0.5

Highest status: Normal

Snapshot omitted: real, identifiable person.

Time in each status: Insufficient evidence 4s,
Normal 1s

Image quality (0–1): 0.48 · **Face visible:** 42% · **Full
body visible:** 40%

Strongest signals: head 0.57, coordination 0.54,
sway 0.00, posture 0.00, eyes 0.00

Main evidence shown: Head instability

Main limitations shown: Face not visible, Lower
body not visible — gait unavailable, Too few visible
signals

Short slow-motion clip, with only about 5 s of usable
footage. Its result carries little weight.

What was verified in the platform

Area	How verified	Result
AI service health and deployed build	Health check, and the running container inspected for the latest fixes	Pass
Person detection, tracking, body pose, face signals	10-clip replay through the deployed container	Working
Risk engine: scoring, persistence, abstaining	74 unit tests plus the replay	Pass
Safety events: stored, escalated, closed at end of call	Automated API tests	Pass
Reviewer workflow and audit trail	Automated API tests: outcome, breath-test result, mandatory note on confirmation	Pass
Access control	Automated API tests: division staff see only their own division; lobby and crew users get no safety data	Pass
Live alerts to monitors	Automated API tests: alerts published, and delivered only to authorised staff	Pass

Licensing: feature turned off per organisation	Automated API tests: no events stored, indicators hidden	Pass
Live dashboard with a real lobby camera	Not part of this run; the web application was not running during the test	Not tested in this run

Improvements made after the first test round

Issue found	Fix	Measured effect
Wide video made sideways movement look larger than it was	Body measurements corrected for the video's aspect ratio	Sway readings now consistent across camera formats
A person walking toward the camera was read as swaying on the spot	Walking detection also recognises the person growing larger in frame	Kobuleti walk: Monitoring for 13.4 s before, Normal now
Tiny, distant people produced noisy sway and gait readings	Sway and gait are skipped when a person fills less than 20% of the frame height	Fewer spurious signals on wide shots

The impairment-clip results did not change after the fixes: the same clip alerted before and after, with its first alert at 52 s before and 50 s now.

Limitations of this test

- **Small sample.** 10 clips is enough to show the system works and to spot its failure modes. It is not enough to quote accuracy percentages with confidence.
- **The footage is not lobby footage.** Old films, dash-cams and body-cams have different angles, lighting and motion from a fixed lobby camera. Real lobby performance is still unknown.
- **No confirmed ground truth.** The real clips show *suspected* impaired drivers. There are no breath-test results for them.
- **Staged films exaggerate behaviour.** The Chaplin detection shows the logic works on clear cases, not that subtle impairment will be caught.
- **Known weakness:** a person walking slowly straight away from the camera can be partly read as sway, as in the walking-sands clip, which reached Monitoring but no alert.
- **Processing speed:** during this run the container took 101–236 ms per frame on CPU inside Docker (a separate native run took 61–104 ms). That supports roughly one live call at 5 frames per second. Several simultaneous calls will need a GPU or a dedicated server.

Recommendations and next steps

1. **Supervised pilot in 1–2 lobbies** with a fixed camera placed to see the full body and the face, with good lighting. The test shows camera placement matters more than anything else.
2. **Collect ground truth through the built-in review form.** Every alert already records the reviewer's outcome and breath-test result, so the pilot builds a labelled dataset automatically.
3. **Calibrate thresholds on pilot data.** For example, strong eye-closure plus head-instability cases (like the DUI booking clip) may justify a faster alert. That change should come from data, not from these 10 clips.

4. **Before the pilot:** sign off the privacy notice and consent wording, and set a data-retention period for snapshots and events.
5. **Plan hardware for scale:** a GPU server before rolling out beyond a few simultaneous calls.
6. **Keep a human in the loop.** An alert is a prompt to check, never an automatic decision about duty.

Sources

All clips come from Wikimedia Commons. They are public domain, except *Inmate walking in shackles* (CC0), *Walking in the sands* (CC BY 3.0) and *Kobuleti 2025 walking* (CC BY 4.0): *Dream of a Rarebit Fiend* (1906); *One A.M.* (1916, Charlie Chaplin); *Suspected Drunken Driver Confused About Being Arrested...* (police recording, two segments); *Fake Cop Handcuffs Drunk Man...* (Atlanta police body-cam); *Inmate walking in shackles*; *Walking in the sands*; *Kobuleti 2025 walking*; *Deb Haaland and Isleta Pueblo Chief Judge Verna Teller walking*; *Kamala Harris and Joe Biden walking at the White House (slow motion)*. Each clip's source link appears in its detail card above.

Processing cost during this run: 101–236 ms per frame on CPU.